

Gas Water Heater Pilot Light Repair Guide: Fix It Yourself and Avoid a Plumber

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Introduction

No hot water is a different kind of home emergency. It's not dramatic like a burst pipe — but it's arguably more disruptive. No showers. No dishwasher running. No laundry. And the quote from the plumber sits in your inbox: \$247 for the service call, plus parts, plus labor.

What if the actual problem is a \$15 part and 30 minutes of your time?

This guide walks you through diagnosing and fixing the most common reason a gas water heater pilot light goes out — the thermocouple. It also covers when the problem is something else, when you should call a plumber despite your best efforts, and how to make the repair-or-replace decision with confidence.

The core message: You can probably fix this yourself. And even if you can't, understanding what's wrong means you won't get scammed.

Understanding Your Gas Water Heater

Before you touch anything, take 60 seconds to understand what you're working with.

How a Gas Water Heater Works

Cold water enters the tank from the top. A gas burner at the bottom heats the water. The hot water rises and exits through a pipe at the top of the tank. A thermostat controls when the burner turns on and off. A thermocouple is a safety device: it detects whether the pilot light is lit, and if it is, it keeps the gas valve open so the main burner can ignite.

If the thermocouple doesn't detect a lit pilot, it shuts the gas valve. That's the safety function. And that safety function is why a faulty thermocouple is the most common cause of a pilot light that won't stay lit.

Why the Pilot Light Specifically Matters

The main burner doesn't ignite on its own — it needs the pilot light to be burning first. The pilot is a small, continuously burning flame that heats the thermocouple. If the pilot goes out, the thermocouple cools, and the gas valve closes. No pilot, no main burner. No main burner, no hot water.

What You'll See When the Pilot Is Out

- No hot water (or lukewarm water that runs out quickly)
 - The water heater's status light (if it has one) may be off or blinking
 - You may smell a faint gas smell near the water heater — this is normal when the pilot is out and residual gas is in the line
 - The gas control knob may show a indicator (depending on the model)
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The 5-Minute Diagnostic Sequence

Before buying anything, confirm what's actually wrong. This sequence takes 5 minutes and costs nothing.

Step 1: Check the Obvious

This sounds obvious, but plumbers have stories.

- Is the **gas valve** fully open? (The handle should be parallel to the pipe, not perpendicular)
- Is the water heater's **control knob** turned to "pilot" and not "off"?
- Some water heaters have a **safety lockout** that requires the knob to be held down during lighting — read the label near the knob for your model's specific instructions

- Is there a **flashing light** on the water heater? If so, look up the error code on the manufacturer's website

Step 2: Check for a Downdraft

Wind or drafts can blow the pilot flame out. Go outside and check the **flu pipe** (the metal pipe exiting the top of the water heater going upward).

- Is wind coming down the flu pipe?
- Is the **draft hood** (the metal bell shape at the top of the water heater) blocked or damaged?
- Are there any exhaust fans, bathroom vents, dryers, or range hoods running that could be creating **negative pressure** in the house?

If you're in a two-story home and the water heater is on the first floor, running the bathroom fan or kitchen range hood can create enough negative pressure to pull the pilot flame sideways and extinguish it.

Step 3: The Smell Test

- Do you smell gas near the water heater when the control knob is at "pilot"?
- Do you smell gas at the connection points on the gas line?
- If yes to either: **stop and call your gas company or a licensed plumber immediately**. Do not attempt to repair a gas leak. Do not smoke near the area.

A faint gas smell when the pilot is out is normal — it's residual gas in the line. A persistent or strong gas smell when the knob is in any position other than "off" is not normal.

Step 4: The Relight Protocol

Follow these steps exactly — deviations are why most relighting attempts fail.

1. Turn the control knob to **"off"**
2. Wait **5 full minutes** for any residual gas to dissipate
3. Turn the knob to **"pilot"**
4. Push down on the pilot/reset button (or use a tool to hold it if the button is a twist-type knob)
5. While holding the button down, use a **long-match or butane fireplace lighter** — not a regular lighter — to light the pilot port
6. **Keep the button pressed for 60 seconds after lighting** — this is the step most people skip. The thermocouple needs a full minute of heating before it will hold the gas valve open.

7. Release the button slowly
8. Observe: does the pilot stay lit?

If the pilot goes out immediately: The thermocouple is almost certainly faulty. This is the expected outcome — proceed to the repair section.

If the pilot stays lit but the main burner doesn't ignite: This is a different problem — possibly the main burner orifice, gas valve, or thermostat. Call a plumber for diagnostic assessment.

If the pilot goes out after 5-10 minutes of running normally: Could be a weak thermocouple, a draft issue, or a gas supply problem.

The Thermocouple Repair

If the diagnostic sequence pointed to the thermocouple, this is the repair. The thermocouple is the most common failure point in a gas water heater, and replacement is straightforward.

What to Buy

- **Universal thermocouple:** \$8-15 at Home Depot, Lowe's, Ace Hardware, or online
- Bring your water heater's **brand and model number** to confirm fit, or buy a universal kit (universal thermocouples work for 95% of residential gas water heaters)
- **No need for brand-name parts** — a generic replacement works fine
- **Also buy:** a can of **compressed air** or a bottle of **rubbing alcohol and cotton swabs** — for cleaning the pilot orifice while you have everything open

Cost: \$8-15 and 30-45 minutes of your time vs. \$150-400 for a plumber service call.

Tools Needed

- Adjustable wrench (or basin wrench if the water heater is in a tight closet space)
- Flathead or Phillips screwdriver (depending on your model)
- Possibly a 1/4" nut driver
- Flashlight or headlamp

Safety Checklist Before You Start

- Turn the **gas valve to OFF** at the unit

- The gas remaining in the line is minimal and not dangerous, but work in a ventilated area
- No smoking while working on a gas appliance
- Have a helper nearby in case something goes wrong

Step-by-Step Replacement

1. **Remove the access cover** at the bottom of the water heater — this is usually held by two screws or clips
2. **Locate the thermocouple** — it's a thin copper tube (about the diameter of a pencil) with a small tip that sits in or immediately adjacent to the pilot flame
3. **Disconnect the thermocouple from the gas control valve** — the connector is either a push-in fitting or a screw-on fitting. If push-in: pull firmly. If screw-on: unscrew. No tools needed for the connector.
4. **Remove the thermocouple from the pilot assembly** — it's usually held by a small clip or screw near the pilot. Remove the clip and pull the thermocouple out.
5. **Clean the pilot orifice** — use compressed air to blow out the small hole where the pilot gas comes out, or clean it with a cotton swab dipped in rubbing alcohol. Let it dry.
6. **Install the new thermocouple** — insert the new thermocouple into the same position as the old one. The **tip must be positioned in the pilot flame zone** — the flame should wrap around the tip when lit.
7. **Reconnect to the gas valve** — push or screw the connector back in firmly.
8. **Turn gas back on** at the valve.
9. **Relight the pilot** using the exact protocol from the diagnostic section above.
10. **Wait 2-3 minutes** and confirm the main burner ignites and stays lit.

The most common mistake: The new thermocouple tip is not positioned correctly in the flame. The flame must visibly wrap around the tip. If it's too far away (even slightly), it won't generate enough voltage to keep the gas valve open, and the pilot will go out within seconds of releasing the button.

Testing After Repair

- Confirm the main burner ignites when the thermostat calls for heat
- Confirm the main burner stays lit
- Listen for the burner to cycle on and off normally
- Run a hot water tap for 2 minutes to confirm hot water is flowing
- Check the connection points for any smell of gas

When the Problem Is Not the Thermocouple

A few other issues can mimic a thermocouple failure.

Dirty or Clogged Pilot Orifice

If the pilot flame is very small (not fully engulfing the thermocouple tip) or has a yellow/orange color instead of blue, the pilot orifice may be clogged with dust or debris. Clean it with compressed air or a fine wire (available at hardware stores as "pilot light cleaner").

Gas Supply Issues

If the pilot lights but the flame is weak and keeps going out, or if you hear a hissing sound from the gas line — you may have a gas pressure issue or a partially closed main gas valve. Check the full length of the gas line for a valve that's not fully open. If all valves are open and pressure seems low, call your gas company — this is not a DIY repair.

Faulty Gas Control Valve

If you've replaced the thermocouple and cleaned the orifice and the pilot still won't stay lit, the gas control valve itself may be failing. This is an expensive repair (a new gas valve costs \$150-300) and may make more sense to replace the entire water heater if the unit is old.

Replacement cost vs. water heater age: If your water heater is over 12 years old, replace it. If it's 8-12 years old, get a quote for a new unit vs. repair. If it's under 8 years old, a gas valve replacement (\$200-350 including labor) is usually worth it.

Thermocouple Symptoms Checklist

Use this to confirm the thermocouple is your issue:

Symptom	Likely Cause
Pilot lights but goes out immediately when button is released	Thermocouple
Pilot lights, stays lit for several minutes then goes out	Thermocouple (weakening) or draft
Pilot flame is very small / yellow instead of blue	Clogged orifice
Pilot won't light at all	Gas supply or valve issue
Main burner won't ignite even with pilot running	Thermocouple, gas valve, or thermostat
Gas smell when pilot is off	Gas leak — call gas company immediately

The Repair vs. Replace Decision

This is the question most homeowners get stuck on. Here's a straightforward framework.

Replace the Water Heater If:

- **The unit is over 12-15 years old.** Tank-style water heaters typically last 8-12 years, with 15 being the outer edge. If yours is approaching that age and needs a repair, replacement is usually more economical.
- **You see rust-colored water** when you run the hot tap. This means the tank lining is corroding from the inside. It will get worse.
- **You've already made one significant repair** (e.g., replaced the gas valve or thermocouple) and the unit is over 8 years old. The second repair is a sign the unit is declining.
- **The repair cost is more than 50% of replacement cost.** A new gas water heater installed runs \$800-1,500 depending on size and model. If your repair quote is over \$600, get a replacement quote.

Repair the Thermocouple If:

- The unit is **under 10 years old**
- This is the **first significant repair** needed
- The tank shows no signs of internal corrosion (clear water, no rust smell)
- You want to extend the unit's life for another 3-5 years

The Honest Short Version

If you can flush your own water heater, you can replace a thermocouple. It's the first repair most homeowners do on a water heater, and once you've done it, you'll understand your system well enough to handle future maintenance.

Preventing Future Pilot Light Problems

Once you've fixed the thermocouple, a few maintenance steps will extend your water heater's life and prevent recurrence.

Annual thermocouple check: Once a year, remove the access cover and inspect the thermocouple tip. If it appears corroded or discolored, replace it preventively — \$15 now vs. \$200 later.

Annual tank flush: Attach a garden hose to the drain valve at the bottom of the tank, run the other end to a floor drain, and open the valve. Flush until the water runs clear. This removes sediment that can insulate the tank and cause overheating, which damages the thermocouple and gas valve.

Keep the area around the water heater clean: Dust and debris near the burner can clog the pilot orifice over time.

Quick Reference

Symptoms of a bad thermocouple: - Pilot lights but goes out immediately when the button is released - Pilot stays lit for several minutes then goes out - Main burner won't ignite even with a lit pilot

The fix: - Buy a universal thermocouple (\$8-15) - Disconnect the old one from the gas valve and pilot assembly - Install the new one with the tip in the flame zone - Relight and test

When to replace instead: - Water heater is over 12 years old - You see rust-colored hot water - Repair costs exceed 50% of replacement

Safety rules: - Smell gas? Stop. Call the gas company. - Always turn gas to OFF before working on the unit - Position the thermocouple tip in the pilot flame — not just near it

Next Steps

If your water heater is 8 years old or younger and the pilot won't stay lit, the thermocouple fix is almost certainly your solution. Buy the part for \$8-15, set aside 30 minutes, and do it yourself. You'll have hot water tonight and you'll understand your system in a way that pays dividends every time something unexpected happens in your home.

If your water heater is old, the repair costs are adding up, or you just don't want to handle this yourself — get a licensed plumber for an assessment, not a repair. Pay for the diagnostic visit, not the markup on parts and labor.

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